Supplemental notation contract

This supplement is the authoritative notation contract for Active Fedference. The methods, formalism, results, figures, report schemas, and API documentation use these meanings even when a source paper uses a different symbol. A symbol is not reused for a different mathematical object merely because the objects are both probability vectors. The implementation names in the final column are the canonical names for new code and reports; old names survive only as warned, parity-tested compatibility adapters.

Probability objects and generative-model quantities

States, posteriors, and site factors I

States, posteriors, and site factors I

Symbol	Contractual meaning	Canonical implementation term
s	A hidden categorical state; $s \in \{1, \ldots, n_s\}$ is an index, not a distribution.	`state`
o	An observation/outcome index; $o \in \{1, \ldots, n_o\}$.	`observation`
$q_n(s)$	Agent n's local posterior over the shared latent state after its local update.	`local_posteriors[n]`
$q(s)$	The server consensus posterior over the shared state.	`global_posterior` / `consensus`
$q_{-n}(s)$	The normalized cavity posterior with agent n's site contribution removed.	`cavity(...)`
$t_n(s)$	Agent n's site/factor term in natural-parameter space.	`site_factor`
$m_n(s)$	Bridge-only source-equation log potential: $m_n(s) = \log q_n(s) + \kappa_n$ with state-constant κ_n. It is not a claim that every source-protocol message is a broadcast posterior.	No project API; notation for the qualified bridge only.

Priors, policies, and POMDP quantities I

Priors, policies, and POMDP quantities I

Symbol	Contractual meaning	Canonical implementation term
$\pi_0(s)$	A prior over hidden states. The subscript distinguishes a prior from a policy.	`prior` / `log_prior`
π	A policy or action sequence; it is not a prior and is bold when needed.	`policy`
$A[o, s] = P(o \mid s)$	Observation likelihood matrix; each state-indexed column is a pmf.	`likelihood`
$B[s', s, u] = P(s' \mid s, u)$	State-transition tensor indexed by next state, current state, and control.	`transition`
$C[o]$	Log-preference over outcomes; $p_C(o) = \mathrm{softmax}(C)[o]$ is the preferred-outcome pmf.	`log_preferences`
$D_0[s]$	Initial hidden-state prior in the POMDP.	`initial_prior`
$q(o \mid \pi)$	Policy-conditional predicted outcome distribution in EFE calculations.	`predicted_outcomes`

Priors, policies, and POMDP quantities I

The state index s, posterior $q(s)$, prior $\pi_0(s)$, and policy π must not be conflated. In particular, the policy symbol is never used for a prior, and the prior is never called a policy. The uppercase POMDP tensors A, B, C, D_0 are model objects; they are not posterior factors.

For the qualified relation to Friston et al.'s Eq. 7 (Friston et al. 2024), the shared support is finite, every $q_n(s)$ is positive on it, the Eq. 7 softmax input is represented by the bridge-only $m_n(s)$ above, and the declared weights w_n are fixed rather than functions of the emerging consensus. Under exactly those assumptions, additive κ_n constants cancel under softmax and eq. 6 is the categorical posterior-log-potential specialization of the source message-combination term. It neither reconstructs source message construction, cavity/exclusion policy, scheduling, generative factors, nor the complete source protocol.

Divergences, losses, and scalar controls

Generalized-Bayes and aggregation terms I

Generalized-Bayes and aggregation terms I

Symbol	Contractual meaning	Canonical implementation term
$\mathcal{D}(q\|p)$	A regularizing divergence between distributions, such as KL, reverse KL, or α-Rényi.	`divergence`
$L(s;o)$	Loss evaluated at state s for observation o.	`loss_by_state`
$\tau > 0$	Generalized-Bayes learning rate/temperature multiplying the accumulated loss.	`tau` (`learning_rate` is a warned compatibility alias)
w_n	Non-negative raw/base aggregation weight supplied for local posterior q_n.	`base_weights[n]`
a_n	Raw variational server effective weight before normalization. In the variational rule $0 \leq a_n \leq w_n$.	`raw_effective_weights[n]`
$\widetilde{a}_n = a_n / \sum_m a_m$	Normalized influence weight returned for interpretation and plotting.	`normalized_effective_weights[n]`

The symbols w_n, a_n, and $\tilde{a}_n$ are deliberately distinct. The first is supplied before aggregation, the second is the variational raw server output, and the third is only its normalized influence representation. The server heuristic's reweighting is not a FedGVI client loss and does not inherit a client-side robustness theorem. `robust_aggregate` is a server heuristic with the tested $c = 0$ recovery identity. `variational_aggregate` owns the explicit finite-simplex objective and the raw-weight bound; that bound is not an estimator-level B-robustness theorem.

Robustness, divergences, and loss controls I

Robustness, divergences, and loss controls I

Symbol	Contractual meaning	Canonical implementation term
$c \geq 0$	Server-side robustness coefficient used by the divergence-reweighting rule.	`robustness`
$\lambda > 0$	Entropy weight in the variational server objective and its coordinate updates. The $\lambda \downarrow 0$ endpoint is a separate deterministic tied-argmax rule.	`entropy_weight`
$\alpha > 0$	Rényi divergence order.	`alpha`
$\beta \geq 0$	Density-power loss parameter.	`beta`
$q_{\mathrm{loss}} > 0$	Robust categorical cross-entropy parameter in $L_{q_{\mathrm{loss}}}$. The $q_{\mathrm{loss}} \downarrow 0$ NLL limit is handled separately, and the subscript prevents collision with posterior $q(s)$.	`q_loss`
$\rho \in [0, 1]$	Contamination strength/rate in a declared attack mechanism.	`contamination_rate` / `rate`

For the objective-backed server rule, the complete scalar-control contract is defined for $c > 0$ and $\lambda > 0$:

$$
\begin{aligned}
F_\lambda(q, a) &= \sum_n a_n \, \mathrm{CE}(q, q_n) - \lambda H(q) + \frac{1}{c} \, \mathrm{KL}_{\mathrm{gen}}(a \| w), \\
q &\propto \exp\left(\frac{1}{\lambda} \sum_n a_n \log q_n \right), \\
a_n &= w_n \exp[-c \, \mathrm{CE}(q, q_n)].
\end{aligned} \tag{1}
$$

The implementation uses $\lambda = 1.0$ by default; the `entropy_weight` argument exposes the stated tempered family. The $c = 0$ branch is handled as a recovery limit outside the $c > 0$ objective; at $\lambda = 1.0$ it is the exact project log-linear pool. The $\lambda \downarrow 0$ endpoint is separately implemented as a deterministic tied-argmax rule and is not obtained by substituting $\lambda = 0$ into the displayed objective or update.

Cavity and factor algebra

Cavity and factor algebra I

For a positive-support global posterior and site term, the cavity operation is defined in log space and then normalized:

$$q_{-n}(s) = \frac{q(s)/t_n(s)}{\sum_{s'} q(s')/t_n(s')} = \mathrm{softmax}(\log q(s) - \log t_n(s))\,. \tag{2}$$

The corresponding factor replacement is

$$\log t_n^{\mathrm{new}}(s) = \log t_n^{\mathrm{old}}(s) + \log q^{\mathrm{new}}(s) - \log q^{\mathrm{old}}(s), \qquad t_n^{\mathrm{new}}(s) \leftarrow \frac{\exp(\log t_n^{\mathrm{new}}(s))}{\sum_{s'} \exp(\log t_n^{\mathrm{new}}(s'))}\,. \tag{3}$$

The code-level adapters are `cavity(global_posterior, site_factor)` and `update_factor(old_site_factor, old_global_posterior, new_global_posterior)`. The old keywords `posterior`, `factor`, `old_factor`, `old_posterior`, and `new_posterior` are accepted only with a `DeprecationWarning`; mixed canonical/old calls fail closed. Recombination is tested by normalizing $q_{-n}(s)t_n(s)$ and checking recovery of $q(s)$ to floating-point tolerance. A transported site factor is represented as a pmf, so its arbitrary positive natural-parameter scale is fixed by the explicit normalization above.

Statistical notation and nesting

Statistical notation and nesting I

Symbol	Contractual meaning
n_{seed}	Number of independently seeded worlds/replicates in a declared cell; the inferential unit for seed-level summaries.
n_{trial}	Number of trials nested within one seed and cell; trials are averaged before seed-level inference.
$\Delta = b - a$	Matched robust-minus-naive contrast for the same seed/trial or the declared seed-level reduction.
r_{rb}	Wilcoxon matched-pairs rank-biserial effect, primary standardized effect.
d_{eq}	$2r_{\mathrm{rb}}/\sqrt{1-r_{\mathrm{rb}}^2}$, a secondary rank-biserial-derived display/planning d-equivalent, not raw Cohen's d.
$\mathrm{CI}_{1-\alpha}$	Percentile bootstrap interval for the named estimand, resampling the declared replication unit.
MCSE	Monte Carlo standard error/precision diagnostic for a simulation summary; it is not a confidence interval.
MDE	Observed-design minimum detectable effect diagnostic under its stated approximation; it is not confirmatory evidence.
p, q	Raw p-value and BH-adjusted q-value; the family and ownership are declared with every report.

For the robustness sweep, the primary result is r_{rb} and the matched mean difference $\overline{\Delta}$ with its bootstrap interval. The d-equivalent is retained only as a monotone secondary display and planning input. When $|r_{\mathrm{rb}}| = 1$, the transform diverges; reports use a finite sentinel and captions disclose saturation rather than presenting a million-scale number as a scientifically interpretable effect. Power, prospective sample size, MCSE, and MDE are observed-effect planning/precision diagnostics, not evidence that a confirmatory effect exists.

Statistical notation and nesting I

The predeclared headline display rule is the largest positive r_{rb} among robust methods, with the declared method order as a deterministic tie break. A report must also expose the complete tied-method set, the tie-break, the method with the largest mean $\overline{\Delta}$, and the method with the largest mean at the worst rate. These are distinct summaries; none is a unique scientific winner when the evidence is tied or conditional.

Statistical notation and nesting I

For the review grid, every configured robust method remains an inferential member and a displayed rate-profile curve. No pooled-mean selection creates a curve, interval, or hypothesis-test member for that surface.

Code and manuscript naming map

Code and manuscript naming map I

Retired/ambiguous name	Canonical name	Compatibility rule
`beliefs`, `agent_beliefs`	`local_posteriors`	Warned keyword/property adapters; no silent reinterpretation.
`weights`	`base_weights`	Warned keyword adapter. The federation wire key `agent_weights` remains unchanged.
`agent_weights` result property	`normalized_effective_weights`	Warned property adapter; serialized wire compatibility is preserved.
Variational `agent_weights` argument	`raw_effective_weights`	Warned objective API adapter; reports use the canonical term.
`shared_beliefs`	`shared_posteriors`	Warned diagnostics property adapter.
`loss_vec`	`loss_by_state`	Warned generalized-Bayes keyword adapter.
`cohens_d_from_rank_biserial`	`d_equivalent_from_rank_biserial`	Warned function adapter; the returned value is not raw Cohen's d.
Report `cohens_d`	Report `d_equivalent`	New reports are canonical and schema-versioned; readers must fail closed on an unsupported version.

The wire-level key `agent_weights` is preserved because it is a federation transport contract, not a claim about the scale or meaning of the new result fields. A future wire migration requires an explicit version and a fail-closed reader; it must not silently reinterpret the key.

Source and evidence boundaries

Source and evidence boundaries I

The Friston belief-sharing equations are source equations/protocol claims. Only under the explicit finite-shared-support, posterior-log-potential, and fixed-weight bridge above does the categorical log-linear pool specialize the source message-combination term; the tested $c = 0$ identity remains project-local. The generalized-Bayes and loss limits are implementation analogues checked in the finite categorical model. The contamination, gallery, onset, conditional-world, and review-grid quantities are conditional simulation evidence over declared cells. None of these finite surfaces is an external-data replication, a reconstructed source protocol, a universal attack taxonomy, a causal intervention, or a proof of a server-side robustness guarantee. Open theory, calibration, protocol, continuous-state, external-data, authenticated-federation, and clean-release work remains open in the project TODO and claim-audit documents.